

GENERAL SPECIFICATION G5.40

Low Voltage Variable Frequency Drives

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G5.40 LOW VOLTAGE VARIABLE FREQUENCY DRIVES

G5.40.1 General

(a) Introduction

- (i) This Section applies to all low voltage Variable Frequency Drives (VFD) for the frequency / speed control of motors and motor-driven equipment. Application, rating, speed control, field bus communication and any deviations from this Section will be listed in the motor-driven equipment General Specification G4 and Particular Specification P4. Where such deviations occur, they shall take precedence over this Section.
- (ii) Perform all work; design, supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, and as listed in General Specification G5.2 Standards, Codes and Regulations.

(b) Features

- (i) VFD's shall have in-built energy management features;
- (ii) VFD's shall have in-built asset management and predictive maintenance features; and
- (iii) VFD's shall be IoT-ready.

(c) References

- (i) The following is a list of standards which may be referenced in this Section and the VFD shall meet or exceed all applicable requirements of the following standards, but not limited to the following:
- (ii) Institute of Electrical and Electronics Engineers (IEEE)

Standard	Subject
IEEE 519	Recommended Practices and Requirements for Harmonic Control in Electric Power Systems.

(iii) International Electrotechnical Commission

Standard	Subject
IEC 60034	Rotating Electrical Machines.
IEC 60099	Surge Arresters.
IEC 60252	AC Motor Capacitors.
IEC 60255	Measuring Relays and Protection Equipment.
IEC 60364	Low Voltage Electrical Installations.
IEC 60529	Degrees of Protection Provided by Enclosures (IP Code).

Standard	Subject
IEC 60618	Inductive Voltage Dividers.
IEC 60831	Shunt Power Capacitors of the Self-Healing Type for a.c. Systems Having a Rated Voltage up to and Including 1000V.
IEC 60871	Shunt Capacitors for a.c. Power Systems Having a Rated Voltage Above 1000V.
IEC 60947	Low-Voltage Switchgear and Control Gear.
IEC 61000	Electromagnetic Compatibility (EMC).
IEC 61642	Industrial a.c. Networks affected by Harmonics.
IEC 61643	Surge Protective Devices Connected to Low- Voltage Power Distribution Systems.
IEC 61800	Adjustable Speed Electric Power Drive Systems.

(iv) Spring Singapore (SS)

Standard	Subject
CP 5	Code of Practice for Electrical Installations.
SS 551	Code of Practice for Earthing.
SS 555	Code of Practice for protection against lightning
SS 538	Code of Practice for maintenance of electrical equipment of electrical installations.
SS 358	Polyvinyl Chloride Insulated Cables of Rated Voltages up to and including 450/750V.

(d) Definitions

(i) Terms that may be used in this Section:

- 1) VFD: Variable Frequency Drive;
- 2) CMOS: Complementary Metal Oxide Semiconductor;
- 3) CSI: Current Source Inverter;
- 4) EMU: Energy Monitoring Unit;
- 5) GTO: Gate Turn-Off thyristor;
- 6) MPR: Motor Protection Relay;
- 7) MTBF: Mean Time Between Failure;
- 8) PWM: Pulse Width Modulation;

- 9) ROM: Read Only Memory;
 - 10) RTD: Resistance Temperature Detector;
 - 11) RTU: Remote Terminal Unit;
 - 12) Rated Load: Load specified for the equipment;
 - 13) Rated Speed: Nominal rated (100 percent) speed specified for the equipment;
 - 14) TDD: Total Demand Distortion;
 - 15) THD: Total Harmonic Distortion;
 - 16) TTL: Transistor -Transistor Logic; and
 - 17) IoT: Internet of Things.
- (e) System Description
- (i) Performance Requirements
 - 1) Rated Continuous Operation Capacity: Not less than 1.1 times full load current rating of driven motor, as indicated on the motor nameplates, and suitable for continuous operation at any continuous overload which may be imposed on motor by driven equipment operating over specified speed range.
 - 2) Basis for Harmonic Computations: Using simplified plant single-line diagrams for current and voltage computations, provide active harmonic filters, line reactors, isolation transformers, or higher pulse converter arrangements required to meet current/voltage distortion and line notching limits.
 - 3) Normal and Emergency Source Current Harmonic Distortion:
Compute normal and emergency source individual and total current harmonic distortion at the low side terminals of the step-down transformer or emergency transformer that supply power to the equipment in accordance with the latest edition of IEEE Standard 519.
 - 4) Normal and Emergency Source Voltage Harmonic Distortion: Compute normal and emergency source voltage harmonic distortion at location defined above. All drives shall be low harmonic type or equipped with active harmonic mitigating filtering equipment to meet these THD limits at PCC.
 - 5) Provide isolating transformers or series reactors, active harmonic filters, or other devices necessary for proper system operation. Provide necessary devices and circuits to prevent operation of one drive from adversely affecting operation of other drives supplied from same transformer or same bus.
 - 6) When isolation transformers are used, design to meet K-factor requirements of drive(s) connected.

(ii) Design Requirements

- 1) Design and provide drive system consisting of variable frequency controller, drive motor, certain auxiliary items, and components necessary for complete operating system.
- 2) Other equipment is being powered from same bus as variable frequency drives. Ensure proper operation of drives and other loads under normal and emergency conditions.
- 3) Supply VFDs rated on basis of actual motor full load nameplate current rating.
- 4) Drive System: Convert incoming three-phase, 50-Hz AC power to variable voltage, variable frequency output for adjustable speed operation of a standard AC induction squirrel-cage motor, using the Pulse-Width-Modulation (PWM) technique to produce the variable frequency output.
- 5) The drive system shall use microprocessor based digital control technology to regulate motor operation. The motor current shall closely approximate a sine wave.
- 6) System rated for continuous industrial duty and suitable for use with Standard IEC, inverter duty motors.
- 7) Incoming Line Circuit Breaker: Provide positive means of disconnecting incoming power, and overcurrent protection for the drive system.
- 8) Incoming Line Reactor: Design to minimise harmonic distortion on the incoming power feeder.
- 9) Drives rated 50 kW and below shall be built into the MCC starter module, and drives above 50 kW shall be standalone.

(f) Submittals

(i) Shop Drawings, and Technical Data: As a minimum, submit the following items:

- 1) Overall drive system operating data, including efficiencies, input currents, and power factors, at driven equipment actual load and rated system input voltage, at 0, 40, 60, 80, 100, and 110 percent of rated speed.
- 2) Detailed calculation for individual and total harmonic content (voltage and current) reflected in system normal source supply at driven equipment actual load at 70 and 100 percent of rated speed at locations specified in Single Line Diagrams and load conditions. Normal and emergency source system short-circuit available at drive shall be calculated from data shown on the drawings, emergency generator characteristics and data obtained from the equipment manufacturer under the contract. Use TDD and THD factors as defined in IEEE Standard 519 or equivalent IEC standards to designate total harmonic content.
- 3) VFD output pulse maximum peak voltage, pulse rise time and pulse rate of rise, including any justification for proposed deviation from specified values. Include motor manufacturer's certification that motor insulation will withstand long-term over-voltages caused at motor terminals due to specified output pulse data or any proposed deviation from this data.

- 4) Data on the shelf life of “DC link” capacitor.
- 5) Complete system rating, including all nameplate data, continuous operation load capability throughout speed range of 0 to 120 percent of rated speed.
- 6) Complete variable frequency controller rating co-ordinated with motor full load nameplate current rating; list any controller special features being supplied.
- 7) Controller, reactor, active harmonic filter, and isolating transformer (if applicable) dimensional drawings; information on size and location of space for incoming and outgoing conduit.
- 8) Maximum heat dissipation from enclosure.
- 9) Should separate enclosures and equipment be necessary for filter elements or power factor correction equipment, provide complete dimensional information including location of space for incoming and outgoing conduit, weight, maximum heat loss, and minimum current carrying capacity and recommended wire size for required interconnecting circuits.
- 10) Layout of controller face showing pushbuttons, switches, instruments, indicating lights, etc.
- 11) Complete system operating description.
- 12) Complete system schematic (elementary) wiring diagrams.
- 13) Complete system interconnection diagrams between controller, drive motor, and all related components or controls external to system, including wire numbers and terminal board point identification.
- 14) Single-line diagram of system, including component ratings.
- 15) Description of diagnostic features being provided.
- 16) Descriptive literature for all control devices such as relays, timers, etc.
- 17) Itemised bill-of-materials listing all system components.
- 18) Details on protection system for motors and VFD.
- 19) VFD parameters settings.
- 20) Details on communication system of VFD with SCADA.
- 21) Statement of manufacturer’s qualifications.
- 22) Factory functional test reports.
- 23) Special shipping, storage and protection, and handling instructions.
- 24) Manufacturer’s printed installation instructions.
- 25) Field test reports.
- 26) Manufacturer’s Certification of Proper Installation.
- 27) Manufacturer’s contact details.

- 28) Suggested common and critical spare parts list to maintain equipment in service for a period of 5 years. Include a list of special tools required for checking, testing, parts replacement, and maintenance with current price information.
- 29) List special tools, materials, and supplies supplied with equipment for use prior to and during commissioning and for future maintenance.
- 30) Operation and Maintenance Manual.
- (g) Manufacturer's Qualification
 - (i) Manufacturer: Minimum 10 years' experience in supplying and installing similar size and type variable frequency, controlled speed, drive systems.
 - (ii) Local Service: Manufacturer shall have a well-established service team in Singapore to provide technical support and service.
 - (iii) Manufacturer
 - 1) ISO 9001 certified; and
 - 2) Minimum 10 years of manufacture and development of variable frequency drive systems.
- (h) Spares
 - (i) Provide spare parts, tools and test equipment as specified in General Specification G2.28 Spare Parts, Tools and Test Equipment.

G5.40.2 Products

- (a) Service Conditions
 - (i) Ambient Operating Temperature: 32 to 40 degrees C;
 - (ii) Humidity: 0 to 95 percent relative (non-condensing);
 - (iii) Altitude: <1,000 metres Above Sea Mean Level; and
 - (iv) Supply Voltage
 - 1) 400 volts, three-phase, 50-Hz unless otherwise specified;
 - 2) Nominal voltage plus or minus 10 percent;
 - 3) Frequency Stability: Plus or minus 5 percent of maximum frequency; and
 - 4) VFD designs shall be resistant to transients and voltage dips due to starting of the motor and system short circuits.
- (b) Components
 - (i) Drive Units
 - 1) Incorporate a switching power supply operating from a DC bus, to produce a PWM output waveform simulating a sine wave and providing power loss ride through of 2 milliseconds at full load, full speed.
 - 2) Current-limiting semiconductor fuses for protection of internal power semiconductors.

- 3) Employ a diode bridge rectifier providing a constant displacement power factor of 95 percent, minimum at all operating speeds and loads.
- 4) Use transistors for output section, providing a minimum 96 percent drive efficiency at full speed, full load and 92 percent at 20 percent load.
- 5) Employ DC power discharge circuit so that after removal of input power DC link capacitor voltage level will decay below 50 volts DC within 1 minute after de-energising following IEC or BS Standards. Design DC link capacitor for a MTBF of 5 years.
- 6) Operate with an open circuited output.
- 7) Output Voltage: 0 to 400 volts, three-phase, 0- to 50-Hz, minimum.
- 8) Maximum peak voltage of PWM VFD output pulse of 1,000 volts, with pulse rise time of not less than 2 microseconds, and a maximum rate of rise of 500 volts per microsecond. Should magnitudes of these characteristics be more stressful to motor insulation than specified values, provide insulation systems on the motors suitable for the proposed values.
- 9) Motor Audible Noise Level: When operating throughout speed range of PWM VFD, no more than 3 dBA above that designated in IEC or BS standard, for same motor operated at constant speed with a 50-Hz supply voltage.
- 10) Short-Time Overload Capacity: 120 (minimum) percent of rated load in rms current for 1 minute following full load, full speed operation.
- 11) Starting Torque: 140 percent of rated torque for the start-up duration of 2 seconds.
- 12) Derating for Long or Large Cross-Section Motor Cables: Drive units shall be derated for applications where the length of the motor power cables, or their size, exceeds the corresponding values recommended by the manufacturer.
- 13) Equipment Short-Circuit Rating: Suitable for connection to system with maximum source three-phase, bolted fault, short-circuit available of 65,000 amperes rms symmetrical at 400 volts.
- 14) Supply and install drives with output current-limiting reactors and output filters. Line reactors mounted external to the drive enclosure shall be in a matching enclosure rated IP 21.
- 15) Diagnostics: Comprehensive for drive adjustment and troubleshooting:
 - Memory battery backup; 100-hour minimum during a power loss;
 - Status messages will not stop drive from running but will revert it from starting; and
 - Fault Condition Messages and History: First fault protection function to be activated, ability to store six successive fault occurrences in order. Minimum faults numerically:
 - Overcurrent (time and instantaneous);
 - Overvoltage;

- Undervoltage (DC and AC);
- Overtemperature (drive, motor windings, motor bearing, pump bearing);
- Serial communication fault;
- Short-circuit/ground fault (motor and drive);
- Motor stalled;
- Semiconductor fault;
- Microprocessor fault; and
- Single-phase voltage condition.

16) Drive Protection:

- Fast-acting semiconductor fuses;
- Overcurrent, instantaneous overcurrent trip;
- DC undervoltage protection, 70 percent dropout;
- DC overvoltage protection, 130 percent pickup;
- Overtemperature, drive, inverter, converter, and DC link components;
- Overtemperature, motor, and pump;
- Single-phase protection;
- Reset overcurrent protection (manual or automatic reset);
- Active current limit/torque limit protection;
- Semiconductor fault protection;
- Short-circuit/ground fault protection;
- Serial communication fault protection;
- Microprocessor fault;
- Surge protection for transient overvoltage (6,000 volts, 80 joule surge, tested per IEC 60099; and
- Visual display of specific fault conditions.

17) Operational Features:

- Use manufacturer's standard unless otherwise specified;
- Sustained power loss;
- Momentary power loss;
- Power interruption;
- Power loss ride through (0.1 second);
- Start on the fly;
- Electronic motor overload protection;

- Stall protection;
 - Slip compensation;
 - Automatic restart after power return (ability to enable/disable function);
 - Critical frequency lockout (three selectable points minimum, by 1.5-Hz steps in 10-Hz bands, to prevent resonance of system);
 - Drive maintenance system software for complete programming and diagnostics;
 - Earth fault protection, drive, and motor; and
 - Operate with no motor connected to output terminals.
- (ii) Rectifier: Three-phase, minimum 6 pulse full wave diode bridge rectifier to provide a constant DC voltage to the drive's DC bus.
- (iii) Drive units 200 kW and larger shall be 12-pulse (minimum) unless otherwise approved by the S.O.. Smaller units may be 6-pulse (minimum).
- (iv) Provide series choke and capacitors (DC link filter) on DC bus to reduce ripple in rectifier output and to reduce harmonic distortion reflected into incoming power feeders.
- (v) Controller: Microprocessor-controller PWM inverter to convert to DC voltage to variable voltage, variable frequency three-phase AC output. The output voltage shall vary proportionally with the frequency to maintain a constant ratio of volts to hertz up to 50-Hz. Above 50-Hz, the voltage shall remain constant, with the drive operating in a constant power output mode.
- (vi) Enclosure
- 1) IEC IP 21 (minimum), gasketed, free-standing enclosure or MCC mounting as shown on single-line diagrams, completely front accessible, and hinged doors. Properly sized to dissipate heat generated by controller within limits of specified operating conditions (including ambient temperature and ambient airflow). Enclosure not to exceed dimensions shown on the Drawings.
 - 2) Supply drive complete with cable termination compartment door interlocked main circuit breaker, defeatable (lockable in the open position), emergency stop pushbutton, alphanumeric keypad and display, and operator's controls.
 - 3) Wire drive from below for power and control wiring. Motors cable termination shall be suitable for termination of shielded motor cabling in accordance with drive supplies recommendation.
 - 4) Size forced-ventilation for periodic operation to cool each unit with maximum room ambient temperature of 37 degrees C. Provide redundant fans such that if one fan fails remaining fans supply adequate ventilation for the drive when operating at maximum capacity. Provide air filters on ventilation intakes.

- 5) Bundle stranded copper wiring neatly with nylon tie wraps or with continuous plastic spiral binding; label each terminal for permanent identification of leads; identify each wire at each end with imprinted mylar adhesive-back wire markers; incorporate in as-installed wiring diagrams for wire and terminal numbers shown; wiring across door hinges use 19-strand, stranding looped for proper twist rather than bending at hinge; wire connections internal to panels by crimp-on terminal types. For multiple enclosure systems, complete interconnection wiring with gasketed enclosure openings for wiring; multipoint plug receptacles for any control wiring crossing equipment shipping splits.
 - 6) Selector switches, indicating lights, potentiometers, instruments, protective devices, major system components, etc., identified by means of mechanically attached, engraved, laminated nameplates.
- (vii) Operator Interface
- 1) Controls: Mount drive local control on front door of enclosure and include control switches or membrane type keypad for the following operator functions:
 - Start (when in local mode);
 - Stop (when in local mode);
 - Speed increase (when in local mode);
 - Speed decrease (when in local mode);
 - Parameter mode selection (recall programmed parameters);
 - LOCAL/OFF/REMOTE control selection (in remote, provide for remote RUN command digital input and speed increase/decrease via remote 4 to 20 mA analogue signal);
 - Fault reset, manual for all faults, except loss of AC voltage which is automatic upon return;
 - RUN/pre-set speed;
 - Parameter lock (password or key switch lockout of changes to parameters); and
 - Start disable (key switch or programmed code).
 - 2) Control shall be provided to adjust the VFD for manual operation at MCC (VFD panel) or LCS.
 - 3) Control circuit disconnect shall de-energise circuits in units that are not de-energised by main power disconnect device.
 - 4) Control Voltage: Manufacturer's standard inside of enclosure. 24 volts AC for control circuits external to the drive unit enclosure. Supply power for motor space heaters rated 230 volts.
 - 5) Arrange components and circuits such that failure of any single component cannot cause cascading failure(s) of any other component(s).

- 6) Alphanumeric Display: During normal operation and routine test, the following parameters shall be available:
 - Motor current (percent of drive rated current). Output frequency (Hertz);
 - Output voltage;
 - Running time;
 - Local/remote indicator;
 - Status of digital inputs and outputs;
 - Analogue input and output values;
 - Output motor current per leg; and
 - All test points.
 - 7) Adjustable Parameters: Set drive operating parameters and indicate in a numeric form. Potentiometers may not be used for parameter adjustment. Minimum set-up parameters available:
 - Frequency range, minimum, maximum;
 - Adjustable acceleration/deceleration rate;
 - Volts per Hertz (field weakening point);
 - Active current limit/torque limit, 0 to 140 percent of drive rating;
 - Adjustable voltage boost (IR compensation);
 - Pre-set speed (adjustable, pre-set operating point); and
 - Provision for adjustment of minimum and maximum pump speed to be supplied as function of 4 to 20 mA remote speed signal.
- (viii) Signal Interfaces
- 1) Communication Interface: To suit plant control system
 - As a minimum, the VFD shall be capable of accepting the following commands from a host (master) device:
 - RUN and STOP Command; and
 - Speed signal.
 - As a minimum, provide the following status information to the plant control system:
 - Power supply ON;
 - VFD ON;
 - Drive (motor) speed;
 - Motor real power;
 - Drive FAULT; and
 - REMOTE control selected (Drive ready).

- Provide a minimum of two user defined inputs. The status of the inputs to be capable of being queried by the host.
- 2) Discrete Inputs:
 - Accept discrete inputs from local control station (LCS) located near the motor for REMOTE control selection and START and STOP when REMOTE is selected;
 - Accept discrete EMERGENCY STOP contact operation (fail safe contact) from LCS;
 - Accept MOTOR HIGH TEMPERATURE signal from thermal devices locate in motors; and
 - Frequency resolution shall be 0.1 percent of base speed.
- 3) PTC Thermistor Input: Monitor one PTC thermistor circuit.
- 4) Potentiometer Input: Monitor potentiometer input when REMOTE is selected at the LCS.
- 5) Speed limit pre-setting potentiometers or programmable controls shall be provided to enable the minimum and maximum speeds to be set. The setting shall determine the variable range within which the speed of the motor can be controlled.
- 6) Discrete Outputs:
 - Motor ON (DRIVE RUNNING) contact output for status indication at LCS;
 - DRIVE FAULT (with common contact closure for all fault conditions); and
 - DRIVE IN REMOTE MODE.
- 7) Analog Input: When LOCAL/OFF/REMOTE switch is in REMOTE, control drive speed from remote 4 to 20 mA DC signal:
 - Make provisions for adjustment of minimum and maximum motor speed which shall result from this signal;
 - Factory set this adjustment to comply with operating speed range designated in the driven equipment Specifications G4 and P4; and
 - Frequency resolution shall be 0.1 percent of base speed.
- 8) Analog Output: Provide two 4 to 20 mA DC signals for actual frequency, actual load.
- (ix) Accessories
 - 1) Equipment Identification Plate: 1.5mm stainless steel with 6mm die-stamped equipment tag number securely mounted in a readily visible location;
 - 2) Lifting Lugs: Equipment weighing over 45kg;
 - 3) Anchor Bolts: BS EN 10088 Type 316 stainless steel, sized by equipment manufacturer; and
 - 4) Communication Interface: RS 485 compatible with PROFIBUS DP along with necessary cables.

- (c) Factory Finishing
 - (i) Enclosure
 - 1) Primer: One coat of rust-inhibiting coating.
 - 2) Finish:
 - Interior: One coat of manufacturer's standard corrosion-resistant enamel;
 - Exterior: One coat manufacturer's standard corrosion-resistant, fungicidal paint; and
 - Grey, BS 5252 Colour Code 00.A.05, gloss finish.
 - (ii) Conformal coating and tropicalisation of the electronic components shall be provided.
- (d) Testing
 - (i) Factory Inspections: Inspect control panels for required construction, electrical connection, and intended function.
 - (ii) Factory Tests and Adjustments: Test all control panels supplied.
 - (iii) Record test data for report.
 - (iv) Functional Test: Perform following tests as a minimum:
 - 1) Test diodes, transistors, and GTOs at a thermal level of 125 degrees C;
 - 2) Test TTL and CMOS chips at 70 degrees C;
 - 3) Test printed circuit boards while heat cycled to maximum temperature of 65 degrees C;
 - 4) Test run power sections at maximum 40 degrees C for 12 hours and run with motors for 6 hours;
 - 5) Test assembled drive at maximum 40 degrees C and full load, full speed for 4 hours;
 - 6) Test power capacitors and active components;
 - 7) Operate controller with motor throughout its specified range, and at rated power supply load for 1 hour;
 - 8) Resonance: When active harmonic filters are supplied to meet specified harmonic distortion requirements, perform analysis and supply documentary evidence that filter elements do not resonate with the remainder of system parameters at any of harmonic frequencies present; and
 - 9) Additional tests which are the standard of the manufacturer.

G5.40.3 Execution

- (a) Installation
 - (i) Install in accordance with manufacturer's printed instructions.

(b) Field Testing

(i) Functional Test

- 1) Conduct on each controller;
- 2) Inspect controller for electrical supply termination connections, interconnections, proper installation, and quiet operation;
- 3) Vibration Test: Complete assembly, consisting of motor, load, and flexible shafting, connected and in normal operation, shall not develop amplitudes of vibration exceeding limits recommended by current edition of Hydraulic Institute Standards. Where loads and drives are separated by intermediate flexible shafting, measure vibration both at top motor bearing and at two points on top pump bearing, 90 degrees apart; and
- 4) Record test data for report.

(ii) Performance Test

- 1) Conduct on each controller;
- 2) Perform under actual or approved simulated operating conditions;
- 3) Test for continuous 12-hour period without malfunction;
- 4) Demonstrate performance by operating the continuous period while varying the application load, as the input conditions allow, to verify system performance;
- 5) Record test data for report; and
- 6) With Plant load connected to normal utility source, measure the following to show parameters within specified limits:
 - Total and individual current harmonic distortion (up to and including 35th harmonic) at locations specified under following load conditions:
 - VFDs running at full load and half load; and
 - Half of the specified VFDs running at full load and half load.
 - Power factor at input side of each drive. Documented verification that power factor is maintained at 90 percent or greater as speed of drive goes down from 100 percent to 33 percent;
 - THD at locations specified under following conditions:
 - VFDs running at full load and half load; and
 - Half of specified VFDs running at full load and half load.

(iii) Test Equipment

- 1) Use Dranetz, Model No. 626-PA, harmonic distortion monitor and Series 626 disturbance analyser or equivalent instrument to document results.
- 2) Provide diagnostic plug-in test card complete with instructions, multi-position selector switch, and metres or built-in diagnostic control panel or ROM-based processor for monitoring AC, DC, and digital signals to assist in troubleshooting and commissioning of drive.

(c) Manufacturer's Services

- (i) Inspection, Commissioning and Site Adjustment: In addition to the requirements of General Specifications, G2.25 Manufacturers' Services and G2.26 – Plant Testing and Commissioning an authorised service representative or training instructor of the manufacturer shall visit Site as required for drive system to perform the following services:
 - 1) Verify proper installation of the equipment;
 - 2) Inspection, checking and adjusting the equipment;
 - 3) Completion of Manufacturer's Certificate of Proper Installation; and
 - 4) Commissioning and field-testing of the VFD controllers.
- (ii) Training of operation and maintenance Personnel: The authorised service representative or training instructor shall train the operation and maintenance personnel in all aspects of drive operation and maintenance, including step-by-step troubleshooting procedures with necessary test equipment and in accordance with General Specification G2.27 Training, Advisory and Technical Support.

END OF SECTION